

Project Progress Report 3

MARI — Motion-Aware Adaptive Range Index

CSE511 • Reporting period: **Weeks 4–5**

Team: **Member A** = Shah Nadim Kamran Rian • **Member B** = Saheeb Tareque

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Recap. Weeks 2–3 delivered the theory (all theorems + the coupling result) and the competitor comparison (MARI 0.024 vs 0.990, all exact). This period covered cross-domain real-data validation and durable-write systems measurement.

1. TODO list for this period

Planned task	Owner	Target
A6. Write dataset rationale + cross-domain validation section	Member A	Week 4
B6. Run the audits and real-data benchmarks; draw drift-CDF figures	Member B	Week 4
A7. Write the durable-write cost-model framing + systems sections	Member A	Week 5
B7. Run <code>native_bench</code> (LMDB/RocksDB/MARI) + durable/contention model	Member B	Week 5

2. What has been done —

Member A [Shah Nadim Kamran Rian]:

- **A6 — Done.** Ran `audit_real/audit_multi` and `real_data_bench/real_multi_bench` on all three datasets. Result: **95–99%** of moves within a modest guard, **0 mismatches**, and the guard beats hard bucketing **3–9×** on real S&P data.
- **A7 — Done.** Implemented/ran `native_bench.py` (real bytes via `/proc/self/io`) and the durable + contention model. Result: MARI writes **58–82×** fewer durable bytes/update than LMDB, in-band with RocksDB.

Member B [Saheeb Tareque]:

- **B6 — Done.** Wrote the cross-domain validation section and the dataset rationale (why S&P/Elo/temps; why not crypto/tick/GPS/social).
- **B7 — Done.** Wrote the durable-write cost-model framing and the systems sections, with honest scoping (Python is indicative, not a cross-system claim).

3. To be done in the next two weeks —

Task	Owner	When
A8. Write the query, crash-recovery, and memory sections (incl. recovery semantics + atomicity caveat)	Member A	Week 6
B8. Run <code>query_bench</code> , <code>recovery_kill</code> (real SIGKILL), <code>memory_bench</code>	Member B	Week 6
A9. Write the robustness & tuning sections + statistical convention	Member A	Week 7
B9. Run <code>tuning/hetero/hotspot/enhanced/gaps</code> benchmarks	Member B	Week 7